Sequences & Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences: finding terms

Find the 12th term of the arithmetic sequence with $a = 5$, $d = 3$.

$a_{12} = 5 + 11(3) = 5 + 33 = 38$.

2 [Foundational] • Target: Arithmetic sequences: finding terms

Write the first four terms of the arithmetic sequence with $a = 7$, $d = -2$.

$7,\ 5,\ 3,\ 1$.

3 [Foundational] • Target: Arithmetic sequences: finding a and d

The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find the first term a and common difference d.

$a + 3d = 17$ and $a + 8d = 42$. Subtracting: $5d = 25 \Rightarrow d = 5$. Then $a = 17 - 15 = 2$.

4 [Foundational] • Target: Arithmetic sequences: finding a and d

An arithmetic sequence has $a_1 = 10$ and $a_6 = 30$. Find the common difference d.

$a_6 = a_1 + 5d = 10 + 5d = 30 \Rightarrow 5d = 20 \Rightarrow d = 4$.

5 [Foundational] • Target: Arithmetic series: the sum formula

Find the sum of the first 20 terms of the arithmetic sequence with $a = 3$, $d = 4$, using $S_n = \frac{n}{2}(2a + (n-1)d)$.

$S_{20} = \frac{20}{2}(6 + 19(4)) = 10(6 + 76) = 10(82) = 820$.

6 [Foundational] • Target: Arithmetic series: the sum formula

Find the sum of the first 15 terms of the arithmetic sequence with $a = 50$, $d = -3$.

$S_{15} = \frac{15}{2}(100 + 14(-3)) = 7.5(100 - 42) = 7.5(58) = 435$.

7 [Foundational] • Target: Arithmetic series: the sum formula

Find the sum of the arithmetic series $2 + 5 + 8 + \cdots + 50$.

Here $a = 2$, $d = 3$; $50 = 2 + (n-1)3 \Rightarrow n = 17$. $S_{17} = \frac{17}{2}(2 + 50) = 17(26) = 442$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the first 10 positive multiples of 6.

These form an arithmetic sequence with $a = 6$, $d = 6$, $n = 10$.

$S_{10} = \frac{10}{2}(12 + 9(6)) = 5(12 + 54) = 5(66) = 330$.

9 [Intermediate] • Target: Geometric sequences: finding terms

Find the 6th term of the geometric sequence with $a = 3$, $r = 2$.

$a_6 = 3(2)^5 = 3(32) = 96$.

10 [Intermediate] • Target: Geometric sequences: finding terms

Write the first four terms of the geometric sequence with $a = 5$, $r = 3$.

$5,\ 15,\ 45,\ 135$.

11 [Intermediate] • Target: Geometric sequences: finding a and r

The 2nd term of a geometric sequence is 6 and the 5th term is 162. Find r and a.

$ar = 6$ and $ar^4 = 162$. Dividing: $r^3 = 27 \Rightarrow r = 3$. Then $a = 6/3 = 2$.

12 [Intermediate] • Target: Geometric sequences: finding a and r

A geometric sequence has $a_1 = 4$ and $a_4 = 108$. Find r.

$a_4 = a_1 r^3 = 4r^3 = 108 \Rightarrow r^3 = 27 \Rightarrow r = 3$.

13 [Intermediate] • Target: Geometric sequences: finding a and r

A geometric sequence has 3rd term 12 and 6th term 96. Find a and r.

$ar^2 = 12$ and $ar^5 = 96$. Dividing: $r^3 = 8 \Rightarrow r = 2$. Then $a = 12/4 = 3$.

14 [Intermediate] • Target: Geometric series: the finite sum

Find the sum of the first 6 terms of the geometric series with $a = 3$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

$S_6 = \frac{3(2^6 - 1)}{2 - 1} = 3(63) = 189$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Geometric series: the finite sum • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the sum of the first 5 terms of the geometric series with $a = 100$, $r = 0.5$.

$S_5 = \frac{100(1 - 0.5^5)}{1 - 0.5} = \frac{100(0.96875)}{0.5} = 193.75.$

16 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 6.01 Limits & Derivatives

Find the sum of the infinite geometric series with $a = 4$, $r = \frac{1}{2}$.

$S = \frac{a}{1 - r} = \frac{4}{1 - \frac{1}{2}} = 8.$

17 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 7.01 Integration & Applications

Express the repeating decimal $0.333\ldots$ as a fraction using an infinite geometric series with $a = \frac{3}{10}$, $r = \frac{1}{10}$.

$S = \frac{3/10}{1 - 1/10} = \frac{3/10}{9/10} = \frac{3}{9} = \frac{1}{3}.$

18 [Challenging] • Target: Sigma notation • Cross-Lesson: 8.01 Exponentials & Logarithms

Use $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{10} k$.

$\frac{10(11)}{2} = 55.$

19 [Challenging] • Target: Recursive sequences • Cross-Lesson: 9.01 Statistics & Probability

A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n + 5$. Find a_5.

$a_1 = 2, a_2 = 7, a_3 = 12, a_4 = 17, a_5 = 22.$

20 [Challenging] • Target: Identifying arithmetic and geometric sequences • Cross-Lesson: 1.01 Algebra & Equations

State whether the sequence $4, 9, 14, 19, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

Arithmetic, with common difference $d = 5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Arithmetic sequences: finding a and d

The 3rd term of an arithmetic sequence is 15 and the 8th term is 45. Find the first term a and common difference d.

a. $a = 3, d = 5$

b. $a = 3, d = 6$ ✓

c. $a = 6, d = 3$

d. $a = 9, d = 6$

$d = \frac{45 - 15}{5} = 6; a = 15 - 2(6) = 3.$

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22 [Foundational] • Target: MCQ — Arithmetic series: the sum formula

Find the sum of the first 30 terms of the arithmetic sequence with $a = 1$, $d = 2$ (the odd numbers).

a. 960

b. 930

c. 870

d. 900 ✓

$S_{30} = \frac{30}{2}(2 + 29(2)) = 15(60) = 900$.

23 [Foundational] • Target: MCQ — Geometric sequences: finding a and r

A geometric sequence has 2nd term 4 and 5th term -108. Find r.

a. $r = 3$

b. $r = -27$

c. $r = -3$ ✓

d. $r = -9$

$r^3 = -108 \div 4 = -27 \Rightarrow r = -3$.

24 [Foundational] • Target: MCQ — Identifying arithmetic and geometric sequences

State whether $6, 11, 16, 21, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

a. **Arithmetic,** $d = 5$ ✓

b. Geometric, $r = \frac{11}{6}$

c. Geometric, $r = 5$

d. Arithmetic, $d = 6$

Each term adds 5: arithmetic, $d = 5$.

25 [Foundational] • Target: MCQ — Identifying arithmetic and geometric sequences

State whether $90, 80, 70, 60, \ldots$ is arithmetic or geometric, and find its 12th term.

a. **Arithmetic,** -20 ✓

b. Arithmetic, -10

c. Arithmetic, 30

d. Geometric, -20

$d = -10$: $T_{12} = 90 + 11(-10) = -20$.

26 [Foundational] • Target: MCQ — Reading a sequence from its terms

The bar chart shows the first five terms of a sequence. Find the common difference.

a. 6

b. 5 ✓

c. 4

d. 9

Each term increases by 5.

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27 [Foundational] • Target: MCQ — Reading a sequence from its terms

Using the same bar chart, predict the 8th term.

a. 29

b. 34

c. 39 ✓

d. 44

$d = 5$, $a = 4$: $T_8 = 4 + 7(5) = 39$.

28 [Foundational] • Target: MCQ — Reading a sequence from its terms

Using the same bar chart, find the sum of the first five terms shown.

a. 70 ✓

b. 66

c. 24

d. 75

$4 + 9 + 14 + 19 + 24 = 70$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Word problems: sequences in context

An employee's starting monthly salary is 3,500 dollars, and it increases by 150 dollars each year. Find the salary in year 6.

a. 3,950 dollars

b. 4,400 dollars

c. 4,100 dollars

d. 4,250 **dollars** ✓

$a = 3{,}500$, $d = 150$: $T_6 = 3{,}500 + 5(150) = 4{,}250$.

30 [Intermediate] • Target: MCQ — Word problems: sequences in context

A bacteria culture starts with 300 cells and doubles every hour. Find the number of cells after 5 hours.

a. 19,200

b. 1,500

c. 9,600 ✓

d. 4,800

$300 \times 2^5 = 300 \times 32 = 9{,}600$.

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31 [Intermediate] • Target: MCQ — Word problems: sequences in context

A theater's ticket sales grow according to a geometric sequence: 200 tickets sold on day 1, with a common ratio of 1.5. Find the number of tickets sold on day 4.

a. 900

b. 675 ✓

c. 300

d. 450

$T_4 = 200(1.5)^3 = 200(3.375) = 675$.

32 [Intermediate] • Target: Word problems: sequences in context

Find the 3rd term of the geometric sequence with $a = -2$, $r = 2$.

a. -8 ✓

b. 8

c. -11

d. -5

$a_3 = -2(2)^2 = -2(4) = -8$.

33 [Intermediate] • Target: MCQ — Arithmetic sequences: finding terms

[illegible] [illegible] [illegible] [illegible] [illegible] n [illegible] [illegible] [illegible] $S_n = 2n^2 + 3n$. [illegible] [illegible] [illegible] [illegible] a_7 [illegible] [illegible] [illegible]. In an arithmetic sequence, the sum of the first n terms is given by the formula $S_n = 2n^2 + 3n$. Find the value of the 7th term, a_7, of this sequence.

a. 70

b. 30

c. **29** ✓

d. 28

[illegible] [illegible] [illegible] [illegible] [illegible] a_n [illegible] [illegible] $a_n = S_n - S_{n-1}$ [illegible] $n > 1$. [illegible] a_7[illegible] [illegible] [illegible] S_7 [illegible] S_6. [illegible] [illegible] S_7 [illegible] $n = 7$ [illegible] [illegible] [illegible]: $S_7 = 2(7)^2 + 3(7)$ $S_7 = 2(49) + 21$ $S_7 = 98 + 21$ $S_7 = 119$ [illegible] [illegible] S_6 [illegible] $n = 6$ [illegible] [illegible] [illegible]: $S_6 = 2(6)^2 + 3(6)$ $S_6 = 2(36) + 18$ $S_6 = 72 + 18$ $S_6 = 90$ [illegible] [illegible] [illegible] $a_7 = S_7 - S_6$: $a_7 = 119 - 90$ $a_7 = 29$ [illegible] [illegible] [illegible] [illegible] a_7 [illegible] 29. In arithmetic sequences, the n-t.

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34 [Intermediate] • Target: MCQ — Arithmetic sequences: finding a and d

[illegible] [illegible] [illegible] [illegible] [illegible] $a_1, a_2, a_3, \ldots$ [illegible] $a_1 > 0$ [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 39[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 1053[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] a_1[illegible] Consider a geometric sequence $a_1, a_2, a_3, \ldots$ where $a_1 > 0$ and the common ratio is positive. If the sum of the first three terms is 39, and the sum of the next three terms is 1053, what is the first term a_1?

a. **3** ✓

b. 6

c. 9

d. 12

[illegible] [illegible] [illegible] $a_n = a_1 r^{n-1}$. Let the geometric sequence be $a_n = a_1 r^{n-1}$. [illegible] [illegible] [illegible] [illegible] [illegible]: The sum of the first three terms is: $S_3 = a_1 + a_2 + a_3 = a_1(1 + r + r^2) = 39 \quad (1)$ [illegible] [illegible] [illegible] [illegible] [illegible]: The sum of the next three terms is: $a_4 + a_5 + a_6 = a_1 r^3 + a_1 r^4 + a_1 r^5 = a_1 r^3 (1 + r + r^2) = 1053 \quad (2)$ [illegible] [illegible] (2) [illegible] [illegible] (1)[illegible] [illegible] [illegible]: Dividing equation (2) by equation (1), we get.

35 [Intermediate] • Target: MCQ — Arithmetic series: the sum formula

[illegible] [illegible] [illegible] [illegible] a [illegible] r. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 12[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 36[illegible] [illegible] [illegible] a. A geometric sequence has first term a and common ratio r. If the sum of the first two terms is 12, and the sum of the second and third terms is 36, find the value of a.

a. 2

b. 3

c. **4** ✓

d. 6

[illegible] [illegible] [illegible] [illegible] $a, ar, ar^2, \ldots$. [illegible] [illegible]: $a + ar = 12 \Rightarrow a(1 + r) = 12$ (1) [illegible] [illegible]: $ar + ar^2 = 36 \Rightarrow ar(1 + r) = 36$ (2) [illegible] [illegible] (2) [illegible] [illegible] (1): $\frac{ar(1+r)}{a(1+r)} = \frac{36}{12}$ $r = 3$ [illegible] [illegible] r [illegible] [illegible] (1): $a(1 + 3) = 12$ $4a = 12$ $a = 3$ We have the geometric sequence $a, ar, ar^2, \ldots$. Equation 1: $a + ar = 12 \Rightarrow a(1 + r) = 12$ (1) Equation 2: $ar + ar^2 = 36 \Rightarrow ar(1 + r) = 36$ (2) Divide Equation (2) by Equa.

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36 [Intermediate] • Target: MCQ — Geometric sequences: finding terms

[illegible] $a_1 = 3$ [illegible] $r = 3$. [illegible] [illegible] 3 [illegible] (S_3)[illegible] A geometric sequence has $a_1 = 3$, $r = 3$. Find the sum of the first 3 terms S_3.

a. 42

b. 38

c. 36

d. **39** ✓

$S_3 = \frac{a_1(r^3 - 1)}{r - 1} = \frac{3((3)^3 - 1)}{3 - 1} = 39$

37 [Intermediate] • Target: MCQ — Geometric sequences: finding a and r

[illegible] $a_1 = 0$ [illegible] $d = -3$. [illegible] [illegible] 5 [illegible] (S_5)[illegible] An arithmetic sequence has $a_1 = 0$, $d = -3$. Find the sum of the first 5 terms S_5.

a. **-30** ✓

b. -24

c. -27

d. -32

$S_5 = \frac{5}{2}(a_1 + a_5) = \frac{5}{2}(0 + (-12)) = -30$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Geometric series: the finite sum • Cross-Lesson: 8.01 Exponentials & Logarithms

[illegible] 10 [illegible] 22. [illegible] a_1 [illegible] d? A given arithmetic sequence has its 3rd term equal to 10 and its 7th term equal to 22. What is the first term a_1 if the common difference is d?

a. **6** ✓

b. 4

c. 8

d. 2

[illegible]: $a_n = a_1 + (n - 1)d$. [illegible]: $a_3 = a_1 + (3 - 1)d = a_1 + 2d = 10$. [illegible]: $a_7 = a_1 + (7 - 1)d = a_1 + 6d = 22$. [illegible] [illegible] [illegible]: (a_1 + 6d) - (a_1 + 2d) = 22 - 10 \) $4d = 12$ $d = 3$. [illegible] [illegible] d \) [illegible] [illegible] a_1: $a_1 + 2(3) = 10$ $a_1 + 6 = 10$ $a_1 = 10 - 6$ $a_1 = 4$. [illegible] [illegible] 4. The arithmetic sequence is given by the formula: a_n = a_1 + (n-1)d \). We have: $a_3 = a_1 + (3 - 1)d = a_1 + 2d = 10$. And.

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39 [Challenging] • Target: MCQ — Infinite geometric series • Cross-Lesson: 9.01 Statistics & Probability

[redacted] $a_n = 5n - 2$. [redacted] [redacted] The arithmetic sequence is defined by the formula $a_n = 5n - 2$. What is the fifth term of this sequence?

a. 18

b. **23** ✓

c. 28

d. 33

[redacted] $a_n = 5n - 2$[redacted] [redacted] $n = 5$ [redacted]: $a_5 = 5(5) - 2$ $a_5 = 25 - 2$ $a_5 = 23$ To find the fifth term of the arithmetic sequence defined by the formula $a_n = 5n - 2$, we substitute $n = 5$ into the formula: $a_5 = 5(5) - 2$ $a_5 = 25 - 2$ $a_5 = 23$

40 [Challenging] • Target: MCQ — Sigma notation • Cross-Lesson: 1.01 Algebra & Equations

[redacted] $\sum_{k=1}^{8} 3(2)^{k-1}$[redacted] Evaluate finite geometric series $\sum_{k=1}^{8} 3(2)^{k-1}$.

a. **765** ✓

b. 768

c. 1530

d. 1536

$a_1 = 3, r = 2, n = 8$. $S_8 = \frac{3(2^8 - 1)}{2 - 1} = 3(256 - 1) = 3(255) = 765$.

41 [Challenging] • Target: MCQ — Recursive sequences • Cross-Lesson: 2.01 Functions & Transformations

[redacted] $\sum_{k=1}^{\infty} ar^{k-1}$ [redacted]: If infinite geometric series diverges, then:

a. $|r| \geq 1$ ✓

b. $|r| < 1$

c. $r > 0$

d. $r < 0$

[redacted] $|r| \geq 1$.

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42 [Challenging] • Target: MCQ — Identifying arithmetic and geometric sequences • Cross-Lesson: 4.01 Trigonometry

[redacted]: $12 + 6 + 3 + 1.5 + \ldots$ [redacted] Sum of infinite geometric series $12 + 6 + 3 + 1.5 + \ldots$?

a. 20

b. 22

c. **24** ✓

d. 26

$a_1 = 12, r = 1/2. S_\infty = \frac{12}{1 - 1/2} = \frac{12}{1/2} = 24.$

43 [Challenging] • Target: MCQ — Reading a sequence from its terms • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. **5** ✓

c. 6

d. 7

The sum of the first n terms of a geometric sequence is given by the formula $S_n = a_1\frac{r^n - 1}{r - 1}$, where a_1 is the first term and r is the common ratio. Given $a_1 = 5$ and $r = 2$, and $S_n = 315$, we can substitute these values into the formula: $315 = 5\frac{2^n - 1}{2 - 1}$ $315 = 5\frac{2^n - 1}{1}$ $315 = 5(2^n - 1)$ Divide both sides by 5: $\frac{315}{5} = 2^n - 1$ $63 = 2^n - 1$ Add 1 to both sides: $63 + 1 = 2^n$ $64 = 2^n$ Since $2^6 = 64$, we have $n = 6$. Let's check the choices.

44 [Challenging] • Target: MCQ — Word problems: sequences in context • Cross-Lesson: 6.01 Limits & Derivatives

[redacted] 10 [redacted] 26. [redacted] [redacted] a_1 [redacted] d [redacted] a_1. In an arithmetic sequence, the third term is 10 and the seventh term is 26. If the first term of the sequence is a_1 and the common difference is d, find the value of a_1.

a. **2** ✓

b. 4

c. 6

d. 8

[redacted] $a_n = a_1 + (n - 1)d$. [redacted] a_3 [redacted] a_7[redacted]: $a_3 = a_1 + (3 - 1)d = a_1 + 2d = 10$ ([redacted] 1) $a_7 = a_1 + (7 - 1)d = a_1 + 6d = 26$ ([redacted] 2) [redacted] [redacted] 1 [redacted] 2 [redacted] d: $(a_1 + 6d) - (a_1 + 2d) = 26 - 10$ $4d = 16$ $d = 4$ [redacted] $d = 4$ [redacted] 1 [redacted] a_1: $a_1 + 2(4) = 10$ $a_1 + 8 = 10$ $a_1 = 10 - 8$ $a_1 = 2$ We know that the formula for the n-th term of an arithmetic sequence is.

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45 [Challenging] • Target: Arithmetic sequences: finding terms • Cross-Lesson: 7.01 Integration & Applications

■■ ■■■■■■■■■■ ■■■■■■■■■ $a_1, a_2, a_3, \ldots$■ ■■■■ ■■■■■ ■■ $a_5 = 23$ ■■■■■■■■ $d = 4$■ ■■■■■■ ■■■■■ a_2. In an arithmetic sequence $a_1, a_2, a_3, \ldots$, if $a_5 = 23$ and the common difference is $d = 4$, find the value of a_2.

a. **11** ✓

b. 15

c. 7

d. 19

■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■ ■■■■■■■■ $a_n = a_1 + (n-1)d$. ■■■■■ ■■ $a_5 = 23$ ■■■■■■■■ $d = 4$. ■■■■■■■■■ ■■■■■ ■■■■■ ■■■■■■ ■■■■■■■■■■ ■■■■■■■■■: $a_5 = a_1 + (5-1)d$ $23 = a_1 + 4(4)$ $23 = a_1 + 16$ $a_1 = 23 - 16$ $a_1 = 7$. ■■■■■■ ■■■■■■■ a_2■ ■■■■■■■ ■■■■ ■■■■■■■: $a_2 = a_1 + (2-1)d$ $a_2 = 7 + (1)(4)$ $a_2 = 7 + 4$ $a_2 = 11$. The arithmetic sequence is given by the formula $a_n = a_1 + (n-1)d$. We know that $a_5 = 23$ and the common difference $d = 4$. Using the general term formula for an arithmetic sequence.

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